



US012398521B2

(12) **United States Patent**
Völker

(10) **Patent No.:** **US 12,398,521 B2**

(45) **Date of Patent:** **Aug. 26, 2025**

(54) **PLATFORM WHEEL DENT**

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(*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 175 days.

(21) Appl. No.: **18/281,012**

(22) PCT Filed: **Mar. 10, 2022**

(86) PCT No.: **PCT/EP2022/056148**

§ 371 (c)(1),

(2) Date: **Sep. 8, 2023**

(87) PCT Pub. No.: **WO2022/189554**

PCT Pub. Date: **Sep. 15, 2022**

(65) **Prior Publication Data**

US 2024/0175220 A1 May 30, 2024

(30) **Foreign Application Priority Data**

Mar. 10, 2021 (GB) 2103275

(51) **Int. Cl.**

E01F 1/00 (2006.01)

B60W 60/00 (2020.01)

(52) **U.S. Cl.**

CPC **E01F 1/00** (2013.01); **B60W 60/0015** (2020.02); **B60W 60/00253** (2020.02)

(58) **Field of Classification Search**

CPC E01C 11/222; G01C 21/34; B65G 69/005
See application file for complete search history.

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Primary Examiner — Hitesh Patel

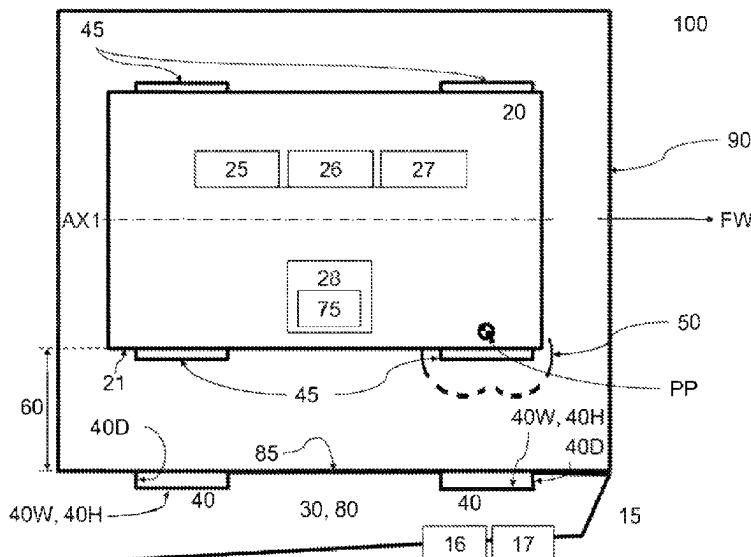
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(57) **ABSTRACT**

A stopping station (15) for use in a transportation network (100) is disclosed. The stopping station (15) comprises a passenger boarding/alighting platform (30) and a plurality of vehicle stopping surfaces (90). The passenger boarding/alighting platform (30) has an upper surface (80) for boarding of passengers into a vehicle (20). The passenger boarding/alighting platform (30) also has a vehicle-facing surface (85) comprising at least two of a cutout (40) for docking of a wheel (45) of the vehicle to the passenger boarding/alighting platform (30). The vehicle stopping surface (90) is adjacent to the vehicle-facing surface (85) and is used for stopping of the vehicle (20) during the boarding/alighting of the passengers in/from the vehicle (20).

15 Claims, 8 Drawing Sheets



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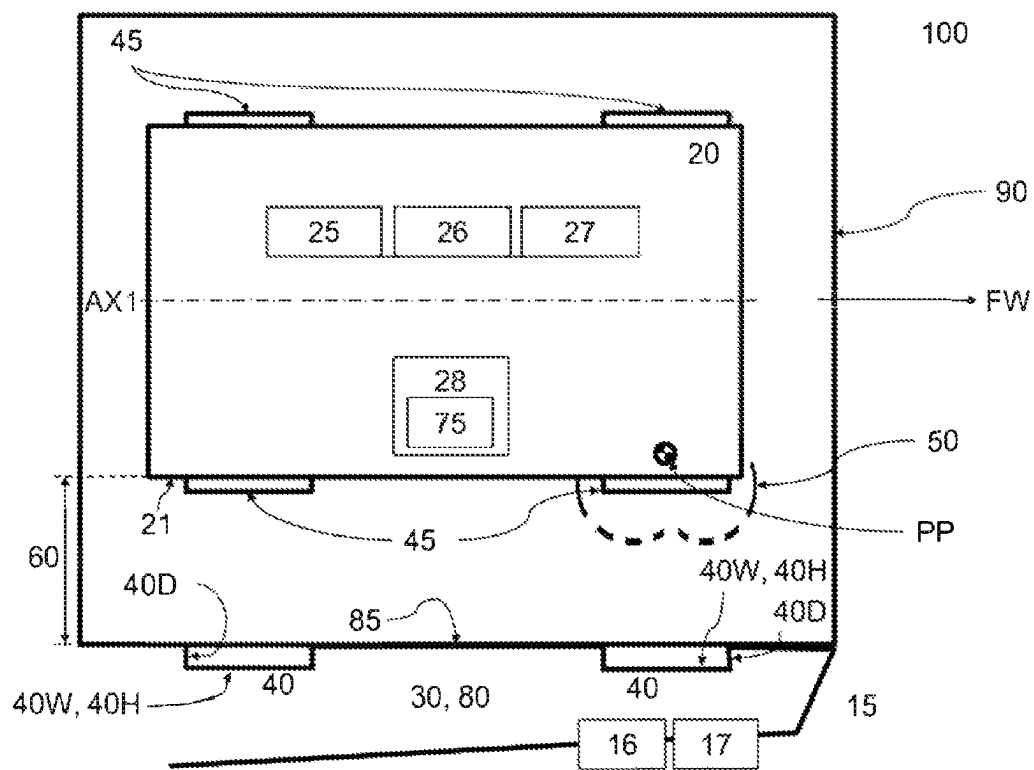
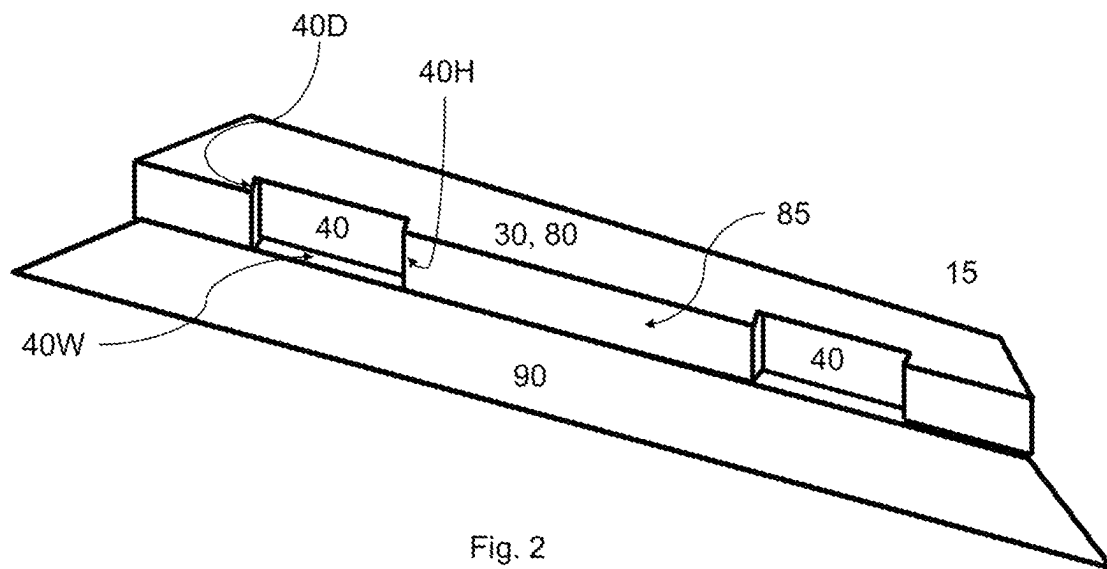


Fig. 1



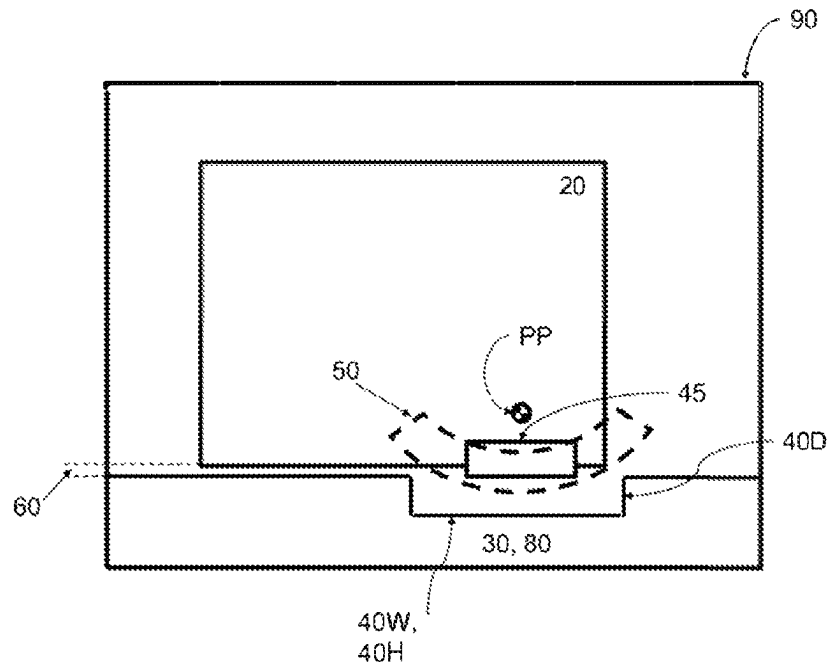


Fig. 3A

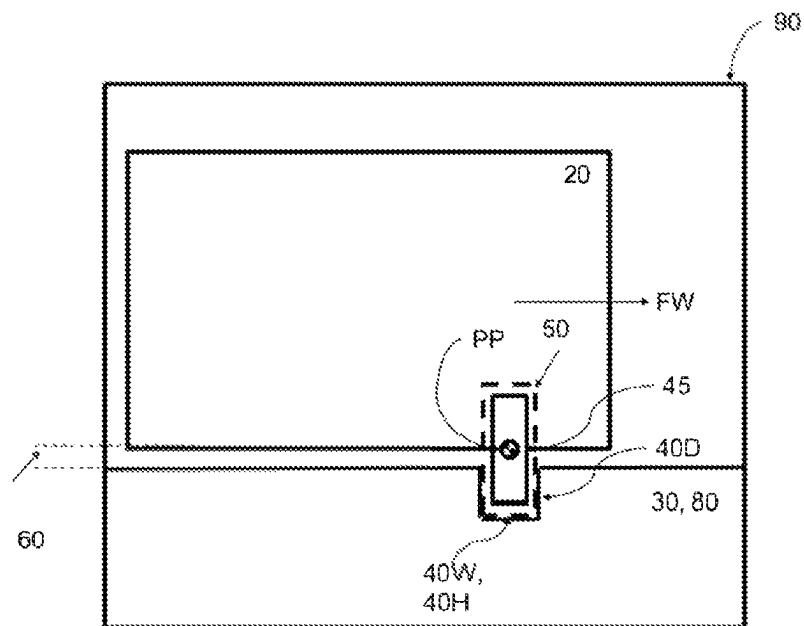


Fig. 3B

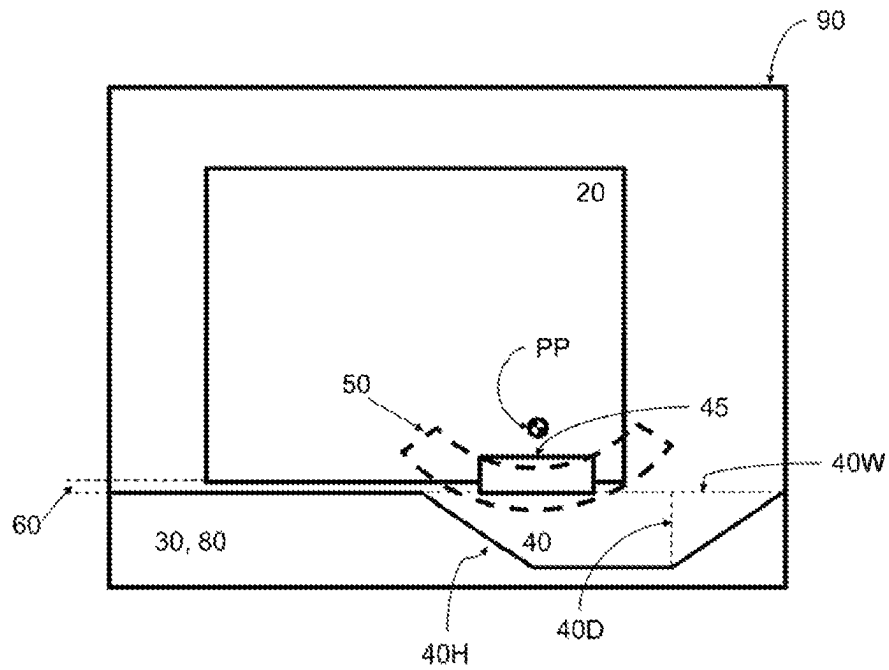


Fig. 3C

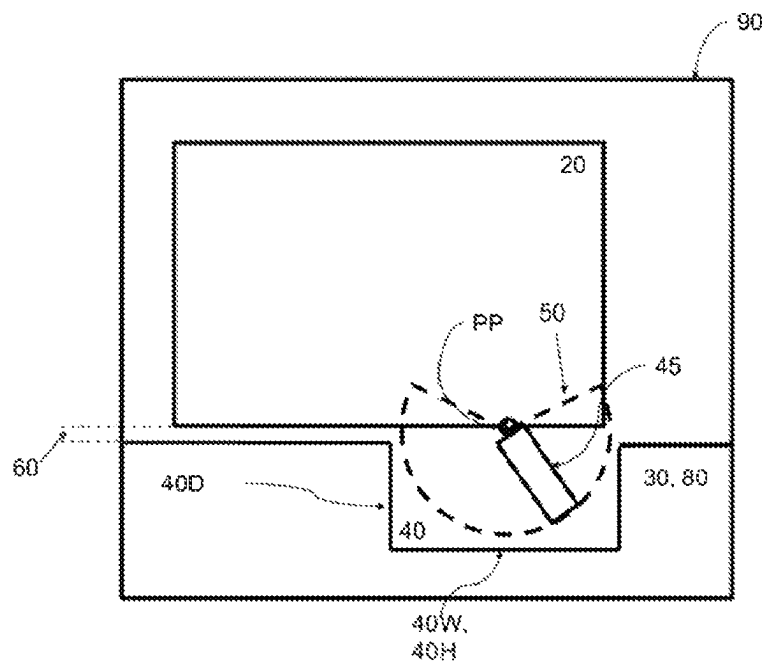


Fig. 3D

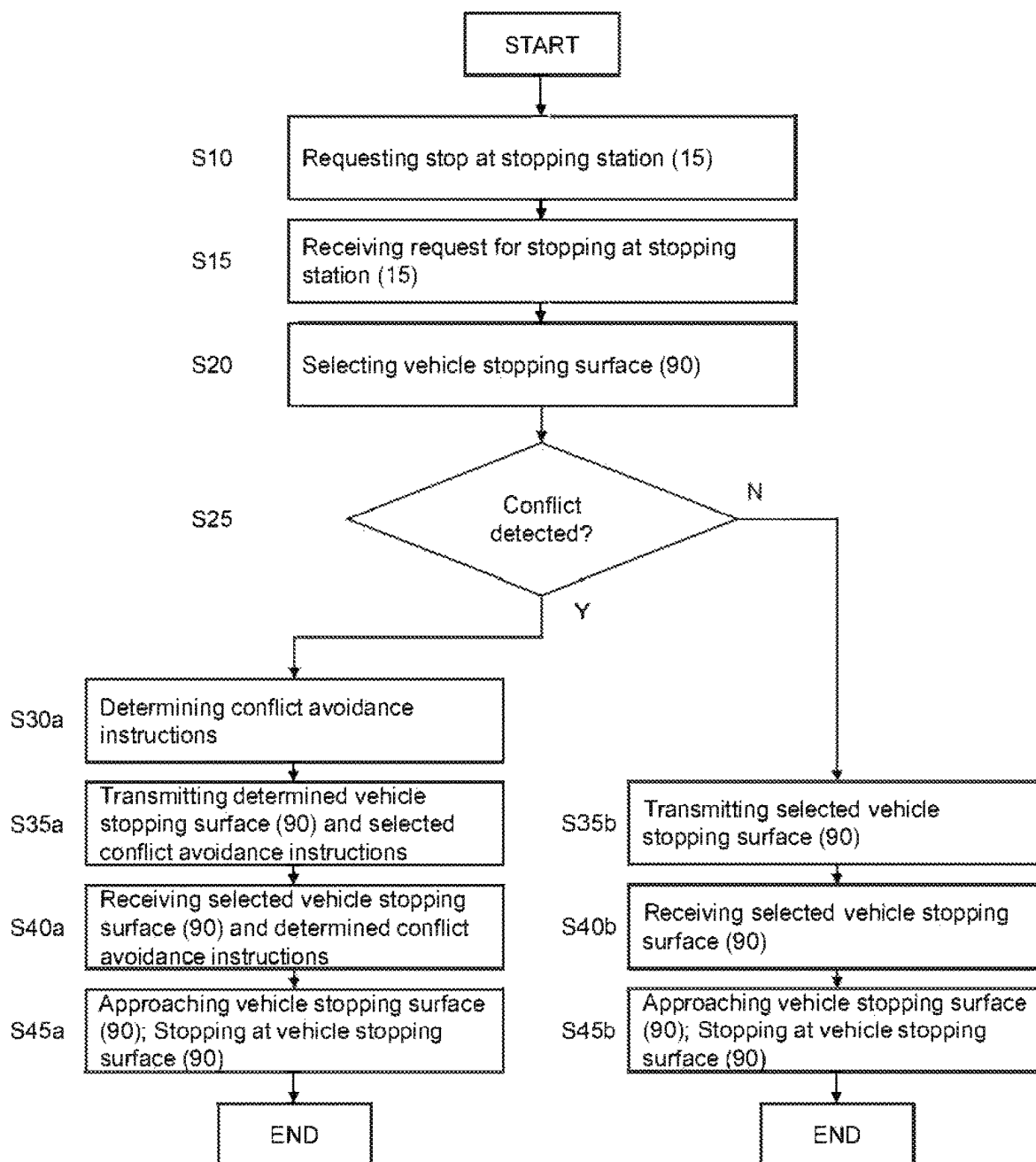


FIG. 4

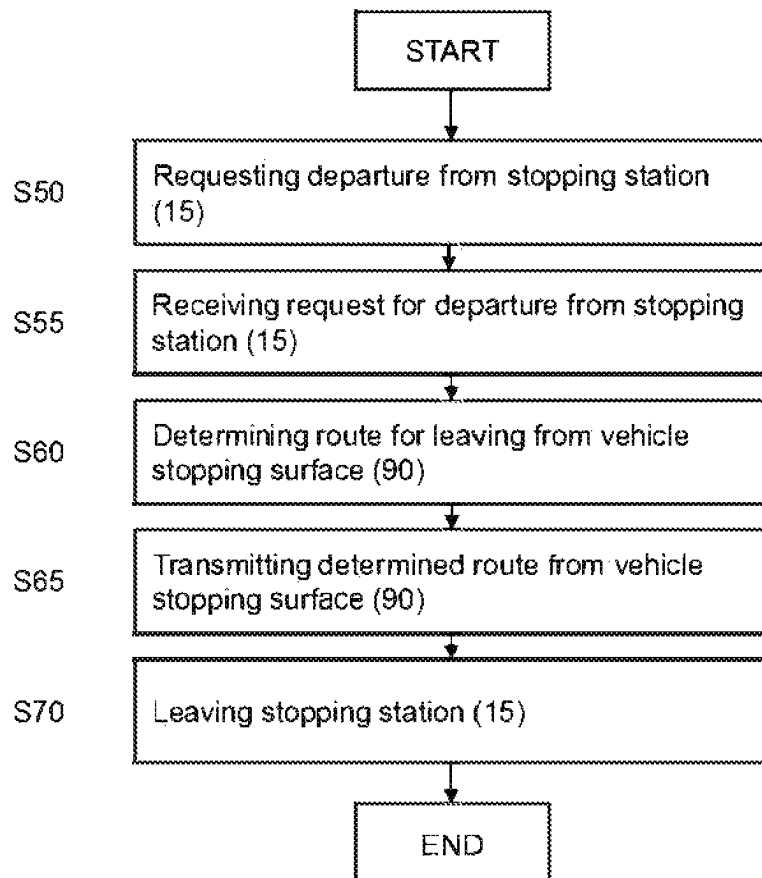


FIG. 5

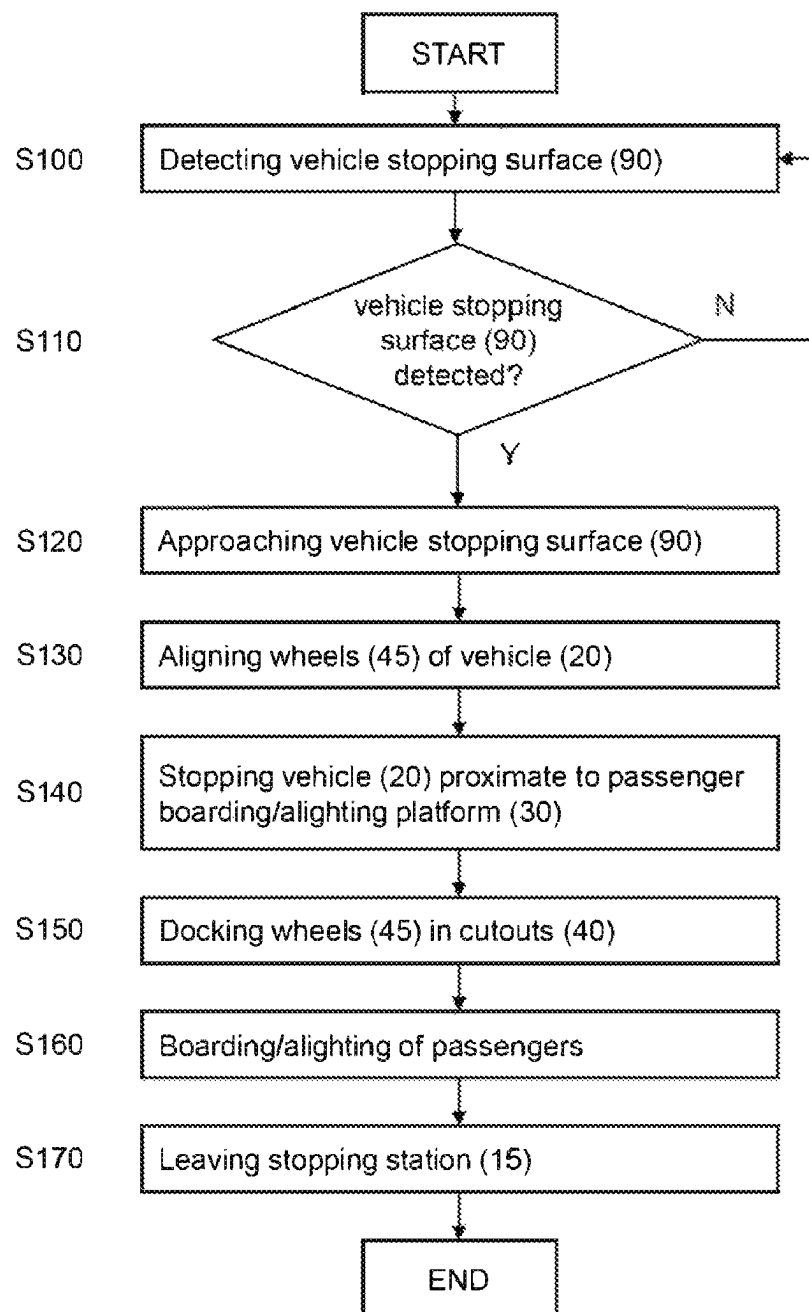


FIG. 6

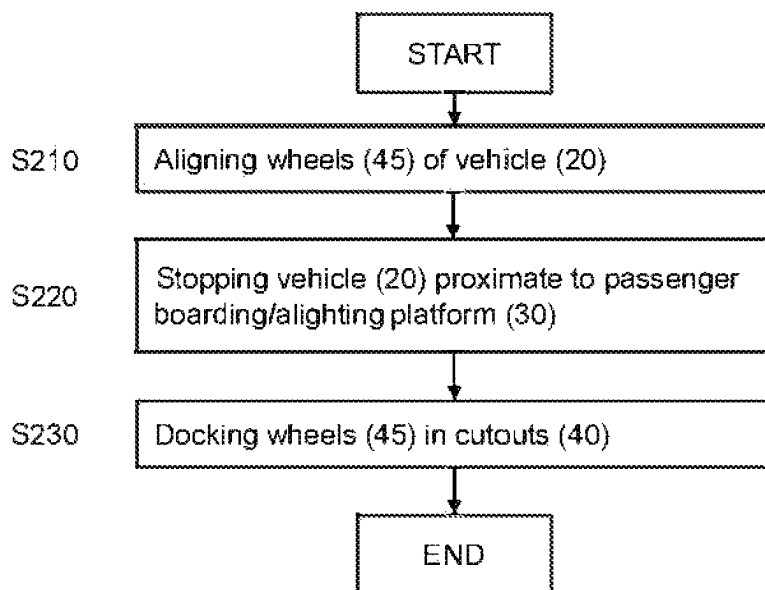


FIG. 7

PLATFORM WHEEL DENT**FIELD OF THE INVENTION**

This application claims priority of the British Patent Application number GB2103275.0, filed on 10 Mar. 2021. The entire disclosure of the British Patent Application number GB2103275.0 is hereby incorporated herein by reference

The field of the invention relates to a passenger boarding/alighting platform having cutouts for stopping of a vehicle close to the passenger boarding/alighting platform and methods for the controlling the movement of the vehicle at the passenger boarding/alighting platform having the cutouts.

BACKGROUND OF THE INVENTION

Transportation networks for transportation of passengers between stopping stations often comprise platforms for boarding/alighting of passengers in/from a vehicle. Depending on the type of platform and the type of vehicle used in the transportation network, a gap between the vehicle and the platform is commonly found in today's transportation networks for allowing the vehicle to approach the stopping station before boarding/alighting passengers or for leaving the stopping station after boarding/alighting of the passengers at the stopping station. The passengers have to cross this gap between the vehicle and the stopping station in order to board/alight in/from the vehicle. Crossing this gap can lead to accidents and/or be a disadvantage for disabled passengers. Different solutions for reducing the gap between the vehicle and the platform are known and used in the current transportation networks around the globe, such as extendible or movable platforms, or "kneeling" vehicles.

MTI Report 12-31 "Automated Transit Networks (ATN): A Review of the State of the Industry and Prospects for the Future, MTI Report 1227". Burford Furman et al. Mineta Transportation Institute Publications (2014) discloses different types of stopping stations as currently used in transportation networks. Multiple concepts for stopping vehicles at the side of a road are disclosed. A system for reducing a gap between the vehicle and the platform is not, however, disclosed in the report.

Self-driving cars or vehicles may be used in the transportation networks. Self-driving cars are often described as being "autonomous", but in practice, there are different classes or levels of vehicle autonomy. The degree of vehicle autonomy is typically divided in five levels, as set out by the On-Road Automated Driving (ORAD) committee of the Society of Automotive Engineers (SAE) in "Taxonomy and definitions for terms related to driving automation systems for on-road motor vehicles" published in Recommended Practice SAE J 3016 on 15 Jun. 2018. Level 0 refers to a vehicle that has no driving automation. The driver of the vehicle is fully in charge of operating the movement of the vehicle. Vehicles of Level 0 may include safety systems such as, for example, a collision avoidance alert. Level 1 refers to vehicles having at least one driving assistance feature such as an acceleration or braking assist system. The driver is responsible for the driving tasks but is supported by the driving assist system which is capable of affecting the movement of the vehicle. Level 2 describes vehicles having more than one assist system for actively affecting the movement of the vehicle. The driver, in Level 2, is still responsible for the driving tasks and must actively monitor the trajectory of the vehicle at all times. The driver is, however, actively supported by the assist systems. Level 3 describes

a so-called "conditional automation" of the vehicle. The vehicle is capable of autonomously driving in certain situations and with limitations. The driver is not required to actively monitor the assist system but is, however, required to take control of a driving situation if requested by the assist system. Level 4 describes autonomously travelling vehicles which are capable of travelling specific routes under normal conditions without human supervision. The vehicles of Level 4 can therefore operate without a driver but might need remote human supervision in case of conflict situations, travelling in remote areas, or when travelling extreme weather conditions. Level 5 Automation describes fully autonomously driving vehicles. No human interaction is required at any time for the operation of the vehicles.

Korean patent application KR 2003/0084874 A (Eom Gi Yeong et al, assigned to KRRI) discloses a safety device installed on a platform to reduce a gap between a vehicle and the platform. The safety device comprises an elastic member that is connected to a fixing member attached to the platform. The elastic member contacts a door of the vehicle when the vehicle is in the station. This contacting of the door of the vehicle helps to close the gap between the platform and the vehicle. The elastic member for the safety device has different degrees of flexibility in a horizontal direction and a vertical direction. The elastic member is flexible in the horizontal direction and absorbs horizontal forces from the vehicle entering the station by deforming the elastic member. The elastic member is not, however, flexible in the vertical direction and is able to withstand a force applied by a passenger standing on the flexible member. A use of cutouts in the platform for the reducing the gap between the vehicle and the platform is not, however, disclosed in the Korean patent application.

Japanese patent application JP H11-209926 A (Tanaka, assigned to Hino Auto Body Ltd) discloses a stopping platform for a vehicle that is formed by notching a sidewalk, wherein the sidewalk is a surface that is a step higher from the road. The sidewalk is notched to fit the length of one vehicle. The notched part is approximately formed to be in the shape of a triangle. Edging stones are arranged at the edge of the sidewalk to distinguish the platform from the road. The Japanese patent application discloses the notching of the sidewalk to accommodate the vehicle during the boarding/alighting of the vehicle. Specific cutouts for the wheels of the vehicle are not, however, disclosed in the patent application.

The prior art discloses solutions for reducing the gap between the vehicle and the platform and rely on reduction of the size of the gap by separate devices that are attached to the platform or the platform is notched to match the size of the vehicles. The prior art does not, however, disclose a system or method for reducing the gap between the vehicle by allowing the vehicle to stop in immediate proximity to the platform.

SUMMARY OF THE INVENTION

The present document describes a stopping station for use in a transportation network. The stopping station comprises a passenger boarding/alighting platform and a vehicle stopping surface adjacent to the passenger boarding/alighting platform. The passenger boarding/alighting platform comprises an upper surface for boarding of passengers in a vehicle and a vehicle-facing surface. The vehicle-facing surface comprises at least two cutouts for docking a wheel of the vehicle to the cutouts of the passenger boarding/alighting platform. The vehicle stopping surface is used for

3

stopping of the vehicle during boarding/alighting of the passengers in/from the vehicle. The form of the cutout is dimensioned according to a wheel hull curve of the wheel of the vehicle and enables rotation of the wheel when the wheel is docked to the cutout. The number of cutouts on a side of the vehicle facing the passenger boarding/alighting platform correlates to the number of wheels of the vehicle. The stopping station of the present document allows, due to the docking of the wheels to the cutouts of the platform, a reduction in the size of the gap between the vehicle and the platform.

The present document further describes a method for the boarding/alighting of the passengers in/from the vehicle at the passenger boarding/alighting platform. The method comprises approaching of the vehicle to the passenger boarding/alighting platform and aligning the wheels of the vehicle with a plurality of cutouts disposed in the vehicle-facing surface of the passenger boarding/alighting platform. The vehicle is caused to stop proximate to the passenger boarding/alighting platform and the wheels are docked to the cutouts of the passenger boarding/alighting platform. The passengers board/alight the vehicle in/from the vehicle and the vehicle subsequently leaves the stopping station by the vehicle. In one aspect of the method, the vehicle detects the stopping station using an onboard positioning unit and an onboard processor of the vehicle.

The present document further describes a method for management of vehicles at the stopping station in the transportation network.

DESCRIPTION OF THE FIGURES

FIG. 1 shows a top view of a stopping station comprising a cutout.

FIG. 2 shows a perspective view of the stopping station comprising the cutout.

FIGS. 3A, 3B, 3C, and 3D show a perspective view of a vehicle stopping at the stopping station.

FIG. 4 shows a flowchart describing a method for controlling a movement of vehicles on a selected one of a plurality of vehicle stopping surfaces at a stopping station in the transportation network.

FIG. 5 shows a flowchart describing a method for controlling a departure of a vehicle requesting to depart from a stopping station.

FIG. 6 shows a flowchart describing a method for boarding/alighting of a passenger at a boarding/alighting platform.

FIG. 7 shows a flowchart describing a method for management of vehicles at the boarding/alighting platform.

DETAILED DESCRIPTION OF THE INVENTION

The invention will now be described on the basis of the figures. It will be understood that the embodiments and aspects of the invention described herein are only examples and do not limit the protective scope of the claims in any way. The invention is defined by the claims and their equivalents. It will be understood that features of one aspect or embodiment of the invention can be combined with a feature of a different aspect or aspects and/or embodiments of the invention.

FIG. 1 shows a top view of a stopping station 15 for use in a transportation network 100 with a plurality of autonomous vehicles 20. The stopping station 15 comprises a passenger boarding/alighting platform 30 and a plurality of vehicle stopping surfaces 90 at which the autonomous

4

vehicles 20 can stop to enable boarding and/or alighting of passengers. The plurality of the vehicle stopping surfaces 90 are, for example, arranged adjacent to each other at the stopping station 15. The stopping station 15 further comprises a stopping station control unit 16 and a stopping station communication unit 17. The stopping station control unit 16 knows the position of the vehicle stopping surfaces 90 within the stopping station 15 and is capable of identifying the individual ones of the vehicle stopping surfaces 90 and the stopping station communications unit 17 communicates with ones of the autonomous vehicles 20 at or near the stopping station 15.

The passenger boarding/alighting platform 30 has an upper surface 80 and a vehicle-facing surface 85. The upper surface 80 is for boarding/alighting of passengers (not shown) into a vehicle 20. The vehicle-facing surface 85 is adjacent to the plurality of the vehicle stopping surfaces 90 and comprises at least two of a cutout 40 for docking of a wheel 45 of the vehicle 20. The transportation network 100 comprises a plurality of stopping stations 15 disposed in the transportation network 100 and connected by roads or tracks (not shown). The transportation network 100 comprises at least one vehicle 20 for travelling between the stopping stations 15 by a passenger (not shown).

The transportation network further comprises at least one roadside control unit (not shown) for controlling of the vehicles 20. The roadside control unit is connected to the stopping station control unit 16 and capable of communicating with the vehicle 20. The roadside control unit is used for controlling of the vehicles 20 when travelling between the stopping stations 15 in the transportation network 100. The roadside control unit will inform the stopping station control unit 16 of approaching ones of the vehicles 20. The stopping station control unit 16 will take over from the roadside control unit a management of the vehicles 20 during approaching and departing of the vehicles 20 to/from the stopping station 15 (as is further elaborated in the description of FIGS. 4 to 7).

The vehicle 20 has a front and a rear. The vehicle 20 has a side 21 and a plurality of wheels 45. The wheels 45 are located, for example, near the front and the rear of the vehicle 20. The wheels 45 located at the front of the vehicle 20 are termed front wheels 45f and the wheels 45 located at the rear are termed rear wheels 45r. The wheels 45 can be flush with the side 21 of the vehicle 20, stick out from the side 21 of the vehicle 20, or can be shifted to be underneath the inside of the vehicle 20. This shifting of the wheels 45 to the inside may, however, have an impact on a stability of the vehicle 20 during maneuvering in curves.

The wheels 45 define a wheel hull curve 50 (the wheel hull curve 50 is only shown on one wheel 45 in FIG. 1 for the sake of clarity of the drawing). The wheel hull curve 50 describes a geometrical outline or surface created by possible positions of an outer surface of the wheel 45 when the wheel 45 rotates and moves during the steering of the wheel 45. More precisely, the wheel hull curve 50 describes, as it is used here, the sum of possible positions of the outer surface of the wheel 45 that the wheel 45 might reach during normal steering operation and during normal driving operation. The wheel hull curves 50 of the wheels 45 of the vehicle 20 might be different and depend on, for example, the position of the wheel 45 relative to the vehicle 20.

The vehicle 20 comprises an onboard processor 27, a vehicle memory 28, a vehicle antenna 25, and a vehicle positioning unit 26. The onboard processor 27 is used for calculating a route to the stopping station 15, and on arrival at the stopping station 15, the onboard processor 27 can also

5

be used to calculate the route to a vehicle stopping surface 90 within the stopping station 15. The vehicle memory 28 is used for storing a structure model 75. The structure model 75 comprises items of data on the transportation network 100 and a position of the stopping stations 15 in the transportation network 100. The vehicle memory 28 is also used for storing further items of data on, for example, the route of the vehicle 20. The vehicle antenna 25 is used for communicating, for example, with beacons disposed in the transportation network 100 or, for example, with the stopping station control unit 16 of the stopping station 15. The onboard positioning unit 26 is used for detecting (see also step S100 described in FIG. 6) the stopping station 15 in the transportation network 100. The vehicle 20 also includes assist systems of Level 2 or Level 3, as described above. The vehicle 20 is, in one aspect, capable of autonomously travelling in the transportation network 100, using the structure model 75 and the onboard processor 27. The vehicle 20 does therefore not require a driver for driving of the vehicle 20.

The wheels 45 can be steered individually or in groups by the onboard processor 27. A first group for the steering of the wheels 45 are the front wheels 45f and a second group for the steering are the rear wheels 45r. The steering of the wheels 45 comprises rotating of the wheels 45 around a pivot point PP (the pivot point PP is only shown on one wheel 45 in FIG. 1 for the sake of clarity of the drawing). This rotating of the wheel 45 for the steering of the wheel 45 comprises changing an orientation of the wheel 45 relative to the direction of a longitudinal center axis AX1 of the vehicle 20 around the pivot point PP. The pivot point PP of the wheel 45 is, when seen from above (as depicted in FIG. 1), located inside of the wheel hull curve 50, inside of the wheel 45, or inside of the vehicle 20.

The wheels 45 are driven individually or in groups by one or several motors (not shown) for powering the vehicle 20. For example, one group are the front wheels 45f and another group are the rear wheels 45r. In a further aspect, only one of the groups of the front wheels 45f or the rear wheels 45r is powered by the motor. The other one of the groups of the front wheels 45f or the rear wheels 45r is not powered by the motor.

The wheels 45 can be attached to the vehicle 20 at a camber angle for improving stability of the vehicle 20 during maneuvering or steering around a curve. The camber angle describes an angle between a vertical axis of the vehicle 20 and a vertical axis of the wheel 45, as viewed from the front of the vehicle 20. A negative camber angle (upper portion of the wheels 45 leaning inward/towards the center axis AX1 of the vehicle 20) can, for example, improve the stability of the vehicle when maneuvering through a sharp curve.

The number of cutouts 40 disposed on the passenger boarding/alighting platform 30 correlates to the number of the wheels 45 on one side of the vehicle 20. The cutout 40 is dimensioned according to the wheel hull curve 50 of at least one wheel 45 of the vehicle 20 and enables rotation of the docked wheel 45 within the cutout 40. A gap 60 between the vehicle 20 and the passenger boarding/alighting platform 30 of the stopping station 15 can be reduced by enabling the wheel 45 to rotate in the docked position. The cutout 40 is dimensioned to have a cutout width 40w, a cutout height 40h, and a cutout depth 40d. The cutout width 40w, the cutout height 40h, and the cutout depth 40d are dimensioned based on the wheel hull curve 50.

The cutout 40 can, in a first aspect, be dimensioned to be similar to the wheel hull curve 50 in shape and size. The

6

wheels 45 will dock closely to the cutout 40 in this first aspect. The cutout 40 can, in a second aspect, also be dimensioned to be similar in shape to the wheel hull curve 50 but to be bigger in size than the wheel hull curve 50. The wheels 45 will dock to the cutout 40 with an offset in this second aspect. The cutout 40 could, in a third aspect, also be dimensioned to be different in shape and/or size from the wheel hull curve 50. The cutout 40 will be dimensioned to accommodate at least the wheel hull curve 50 in this third aspect. The gap 60 is reduced in the first to the third case by allowing the wheel 45 to rotate in the cutout 40.

FIG. 2 shows a perspective view of the stopping station 15 in the transportation network 100. FIG. 2 shows the stopping station 15 comprising cutouts 40 extending from the upper surface 80 of the passenger boarding/alighting platform 30 to the plurality of the vehicle stopping surfaces 90 adjacent to the passenger boarding/alighting platform 30. The cutout height 40w is the same as a vertical distance (step) between the upper surface 80 and the plurality of the vehicle stopping surfaces 90.

FIGS. 3A, 3B, 3C, and 3D show different examples of the cutouts 40 for the docking of the wheels 45 to the passenger boarding/alighting platform 30. The vehicle 20 approaches (step S120 described in FIG. 6) the passenger boarding/alighting platform 30 by driving onto one of the plurality of the vehicle stopping surfaces 90. A driving path of the vehicle 20 during the approaching of the vehicle 20 to the passenger boarding/alighting platform 30 depends on, for example, the number of wheels 45 being steered by the onboard processor 27.

The cutouts 40 are dimensioned according to the wheel hull curve 50 as shown in FIG. 3A. In this first example, the wheel 45 rotates around the pivot point PP within the wheel hull curve 50. The cutout width 40w, the cutout height 40h, and the cutout depth 40d are dimensioned to allow the wheel 45 to rotate around the pivot point PP during the approaching (see step S120) and the docking (see step S150) of the vehicle 20 to the passenger boarding/alighting platform 30. The docking of the wheel 45 to the cutout 40 enables the vehicle 20 to stop proximate to the passenger boarding/alighting platform 30. The gap 60 between the vehicle 20 and the vehicle-facing surface 85 of the passenger boarding/alighting platform 30 is therefore reduced.

FIG. 3B shows a second example of the cutout 40. The vehicle 20 approaches the stopping station 15 by driving sideways in this second example. The wheels 45 are rotated by a 90° steering angle relative to the forward driving direction FW of the vehicle 20. The vehicle 20 can therefore drive sideways for the approaching (see step S120) and the docking (see step S150) of the vehicle 20 to the passenger boarding/alighting platform 30. Setting the wheels 45 to the similar or identical angle relative to the vehicle 20 allows the vehicle 20 to approach the passenger boarding/alighting platform 30 by driving sideways. The driving sideways of the vehicle 20 allows a reduction in the size of the plurality of the vehicle stopping surfaces 90 as the vehicle 20 is capable of maneuvering in a small surface/area. The driving sideways of the vehicle 20 therefore allows the vehicle to approach the passenger boarding/alighting platform 30 in a fast and efficient manner. The driving sideways could also be done by rotating the wheels at angles other than 90° relative to the forward driving direction FW for the approaching of the station 15. The wheels 45 could also be set to 45°, to 60°, or at other angles relative to the forward driving direction FW enabling the driving sideways of the vehicle 20.

In this second example, the cutout width 40w corresponds substantially to the width of the wheel 45. In this second

7

example, the cutout depth **40d** is dimensioned to allow the wheel **45** to dock into the cutout **40** partially or entirely. This docking of the wheels **45** allows the vehicle **20** to stop proximate to the passenger boarding/alighting platform **30**. The gap **60** is therefore also reduced in this second example.

FIG. 3C shows a third example of the cutout **40**. The cutout **40** is dimensioned to be of different shape and size compared to the wheel hull curve **50** in the third example. The cutout **40** of the third example is dimensioned to accommodate the wheel hull curve **50** with an additional space between the wheel **45** and the cutout **40**. The gap **60** between the vehicle **20** and the passenger boarding/alighting platform **30** will be reduced by allowing the wheel **45** to rotate in the cutout **40** when in the docked position.

FIG. 3D shows a fourth example of the cutout **40**. The vehicle **20** approaches the stopping station **15** by driving sideways in a so-called “crab walk” or “dog walk” approach. The crab walk describes independently steering the wheels **45**. The independent steering of the wheels can be achieved by having a four wheel steering system. The independent steering of the wheels **45** allows the wheels **45** to be set to similar or different steering angles relative to the forward driving direction FW and relative to other ones of the wheels **45**. The front wheels **45f** are usually set to steer in a similar or substantially identical direction as the rear wheels **45r**. The front wheels **45f** and the rear wheels **45r** may also be set to different steering angles for turning of the vehicle. This steering of the wheels **45** allows the vehicle **20** to turn efficiently.

The front wheels **45f** are set to steer towards the passenger boarding/alighting platform **30** while the rear wheels **45r** might be set to steer away from the passenger boarding/alighting platform **30** when approaching the passenger boarding/alighting platform **30**. By setting these steering angles for the wheels **45**, the vehicle **20** is able to maneuver swiftly and initiate a turn only requiring a small one of the plurality of vehicle stopping surfaces **90**. In order to approach the passenger boarding/alighting platform **30** more efficiently, the angles of the wheels **45** can be set to change during the approaching of the passenger boarding/alighting platform **30**.

The form or shape of the cutouts **40** is dimensioned to allow the wheels **45** to dock into the cutouts **40** partially or entirely as can be seen from FIGS. 3A to 3D. The wheels **45** on the side of the vehicle **20** facing the passenger boarding/alighting platform **30** and the corresponding wheel hull curve **50** are able to be accommodated within the cutouts **40**. This accommodating of the wheel **45** and/or of the wheel hull curve **50** in the cutouts **40** allows the vehicle **20** to stop proximate to the passenger boarding/alighting platform **30**. The wheels **45** do not interfere with the vehicle-facing surface **85** when docked to the cutouts **40**. The docking thereby reduces the gap **60** (see FIG. 3A to 3D). The vehicle **20** therefore does not need to maintain a distance from the vehicle-facing surface **85** of the passenger boarding/alighting platform **30** for the turning of the wheels **45**. The vehicle **20** can, knowing the exact position of the cutouts **40** in the stopping station **15** and using the onboard positioning unit **26**, stop at the passenger boarding/alighting platform **30** and dock the wheels **45** to the cutouts **40**.

FIG. 4 shows a flowchart describing a method for controlling of the vehicle **20** requesting to stop at the stopping station **15** to one of a plurality of the vehicle stopping surfaces **90**. An approaching vehicle **20** requests to stop at the stopping station **15** in step S10. The requesting is done by sending a request using the vehicle antenna **25** and contains, for example, an expected arrival time of the

8

vehicle **20** at the stopping station **15** or a number of passengers travelling in the vehicle **20**. The stopping station **15** receives the request from the vehicle **20** via the stopping station communication unit **17** in step S15.

The stopping station control unit **16** selects, in step S20, one of the plurality of the vehicle stopping surfaces **90** for the stopping of the requesting vehicle **20**. The selecting of the vehicle stopping surfaces **90** is done, for example, based on the estimated arrival time of the vehicle **20** or the number of passengers travelling in the vehicle **20**. The selecting may also be based on a predicted availability of the vehicle stopping surfaces **90**. The predicted availability can be calculated by the stopping station control unit **16** knowing the approaching vehicles **20** and the number and position of the vehicle stopping surfaces **90**. In other words, the vehicle stopping surfaces **90** are dynamically allocated to the requesting vehicles **20** based on a demand.

The stopping station control unit **16** knows from, for example, the vehicles **20**, the roadside control unit, or sensors disposed in the transportation network **100** or disposed at the stopping station **15**, a position of the vehicles **20** relative to the stopping station **15**. The stopping station control unit **16** also calculates, knowing the position of the vehicles **20**, the route for the requesting vehicle **20** to the selected vehicle stopping surface **90** and detects, in step S25, conflicts between the approaching ones of the vehicles **20**. This calculating of the route to the vehicle stopping surface **90** is done independently from the calculating of the route by the onboard processor **27** of the vehicle **20** (see above). The stopping station control unit **16** knows from this calculation the routes of the plurality of the vehicles **20** approaching the stopping surfaces **90** of the stopping station **15**.

If the stopping station control unit **16** detects, in step S25 the conflict between ones of the approaching vehicles **20** having requested the stopping at the stopping station **15**, the stopping station control unit **16** determines conflict avoidance instructions for avoidance of the determined conflict in step S30a. The selected one of the plurality of the vehicle stopping surfaces **90** and the conflict avoidance instructions are transmitted to the requesting vehicle **20** in step S35a. The requesting vehicle **20** receives items of data identifying the selected vehicle stopping surface **90** and the route to the selected vehicle stopping surface **90** in step S40a. The vehicle **20** approaches (see also step S120 in FIG. 6), using the received items of data on the vehicle stopping surface **90** and the conflict avoidance instructions and stops (see step S140 in FIGS. 6 and S220 in FIG. 6) the vehicle stopping surface **90** in step S45a.

If the stopping station control unit **16** does not detect the conflict between ones of the approaching vehicles **20** having requested to stop at the stopping station **15** in step S25, the selected vehicle stopping surfaces **90** is transmitted to the requesting vehicle **20** in step S35b. The requesting vehicle **20** receives items of data identifying the selected vehicle stopping surface **90** in step S40b. The vehicle **20** approaches (see also step S120 in FIG. 6), using the received items of data on the vehicle stopping surface **90** and stops (see step S140 in FIGS. 6 and S220 in FIG. 6) on the vehicle stopping surface **90** in step S45b.

FIG. 5 shows a flowchart describing a method for controlling a departure of the vehicle **20** requesting to depart from the stopping station **15**. In step S50 the vehicle requests to depart from the stopping station **15**. The requesting to depart from the stopping station **15** by the vehicle **20** is done by a departure request using the vehicle antenna **25** and the stopping station communication unit **17**. The stopping station control unit **16** receives the departure request to depart

from the stopping station **15** by the requesting vehicle **20** in step **S55**. The stopping station control center **16** determines, in step **S60**, a route for the leaving of the stopping station **15** by the requesting vehicle **20**. The determining of the route for the leaving of the stopping station **15** is done using the known position of the vehicles **20** and the known position of the vehicle stopping surfaces **90**. The stopping station communication unit **17** transmits, in step **S65**, the determined route for the leaving of the stopping station **15** to the requesting vehicle **20**. The requesting vehicle **20** leaves, using the route received from the stopping station control center **16**, the stopping station **15** in step **S70**. The stopping station control unit **16** is capable of monitoring the leaving of the vehicle **20** using the sensors. The stopping station control unit **16** monitors, for example, if the passenger or an obstacle is on the vehicle stopping surface **90** or close to the vehicle stopping surface **90**. The stopping station control unit **16** may inform the vehicles **20** of the passenger or the object to avoid a conflict with the vehicles **20**.

FIG. 6 shows a flowchart describing a method for boarding/alighting of the passengers at the passenger boarding/alighting platform **30** into/from the vehicle **20** in the transportation network **100**. The vehicle **20** knows the approximate position of the vehicle stopping surface **90** in the transportation network **100** from the structure model **75** stored in the vehicle memory **28**. The vehicle **20** detects in step **S100**, using the onboard processor **27** and the onboard positioning unit **26**, the exact position of the vehicle stopping surface **90** disposed in the transportation network **100**. A check is conducted in step **S110** to verify that the vehicle **20** has detected the vehicle stopping surface **90**.

If the vehicle **20** has detected the vehicle stopping surface **90** in step **S100**, the vehicle approaches the vehicle stopping surface **90** in step **S120**. If the vehicle has not detected the stopping station, step **S100** is reiterated. Approaching the one of the plurality of the vehicle stopping surfaces **90** is done, by driving from a road of the transportation network **100** onto the one of the plurality of the vehicle stopping surfaces **90**.

The vehicle **20** aligns, in step **S130**, the wheels **45** of the vehicle **20** to the position of the cutouts **40** disposed on the vehicle-facing surface **85** of the passenger boarding/alighting platform **30**. Aligning the wheels **45** of the vehicle **20** to the cutouts of the vehicle-facing surface **85** of the passenger boarding/alighting platform **30** comprises calculating the alignment of the vehicle **20** with the cutouts **15** in the onboard processing unit **26** of the vehicle **20**. The calculating is done using the known position of the stopping station **15** (see step **S100**) and items of information on the position of the cutouts **40** relative to the stopping station **15** stored in the structure model **75**. Aligning the wheels **45** further comprises calculating the angle of the wheels **45** relative to the vehicle **20** during the approaching of the vehicle **20** and setting the angle of the wheels **45** to the calculated angle during the approaching to the passenger boarding/alighting platform **30**. The cutouts **40** are dimensioned according to the wheel hull curve **50** of the vehicle and enable the rotation of the wheel **45**.

The vehicle **20** stops, in step **S140**, proximate to the passenger boarding/alighting platform **30**. The aligning of the wheels **45** to the cutouts **40** disposed in the passenger boarding/alighting platform **30** enables the stopping of the vehicle **20** proximate to the passenger boarding/alighting platform **30**. In a case, where the wheels **45** are flush or aligned with the side **21** of the vehicle **20**, the dimension of the cutouts **40** might only be needed for the turning of the

wheels **45** during the stopping **S140** and the docking **S150** of the vehicle **20** or during the leaving **S170** of the stopping station **15**. In a case, where the wheels **45** stick out from the vehicle **20**, the dimension of the cutouts **40** is, for example, bigger (compared to the dimension of the cutouts when the wheels **45** are flush or aligned with the side **21**).

The vehicle **20** docks the wheels **45** of the vehicle **20** in the cutouts **40** of the passenger boarding/alighting platform **30** in step **S150**. In step **S160**, the passengers board/alight the vehicle **20** from/onto the upper surface **80** of the passenger boarding/alighting platform **30**.

The vehicle **20** leaves the stopping station **15** in step **S170** after the boarding/alighting of the passengers has been completed. The leaving of the stopping station **15** is done in a similar way as the approaching to the stopping station **15** (see step **S120**) above.

FIG. 7 shows a flow chart describing a method for management of the vehicles **20** at the stopping station **15** in the transportation network **100**. The vehicle aligns, in step **S210**, the wheels **45** of the vehicle **20** to the cutouts **40**, as described in step **S130** of FIG. 6 (see above). In step **S220**, the vehicle stops at the stopping station **15** proximate to the passenger boarding/alighting platform **30**. The vehicle **20** docks the wheels **45** in the cutouts **40** in step **S230**.

REFERENCE NUMERALS

- 15** stopping station
- 16** stopping station control unit
- 17** stopping station communication unit
- 20** vehicle
- 21** side
- 25** vehicle antenna
- 26** onboard positioning unit
- 27** onboard processor
- 28** vehicle memory
- 30** passenger boarding/alighting platform
- 40** cutout
- 40_w** cutout width
- 40_h** cutout height
- 40_d** cutout depth
- 45** wheel
- 45_f** front wheel
- 45_r** rear wheel
- 50** wheel hull curve
- 75** structure model
- 80** upper surface
- 85** vehicle-facing surface
- 90** vehicle stopping surface
- 100** transportation network

The invention claimed is:

1. A stopping station for use in a transportation network, the stopping station comprising:
 - a passenger boarding/alighting platform having an upper surface for boarding of passengers into a vehicle;
 - the passenger boarding/alighting platform having a vehicle-facing surface comprising at least two of a cutout for docking of a wheel, wherein the form of the cutout is dimensioned according to a wheel hull curve of at least one wheel of the vehicle and the number of cutouts disposed on the passenger boarding platform correlates to the number of wheels of the vehicle on a side of the vehicle facing the passenger boarding/alighting platform; and
 - a plurality of vehicle stopping surfaces adjacent to the vehicle-facing surface, wherein the plurality of the

11

vehicle stopping surfaces are used for the stopping of the vehicle during boarding/alighting of the passengers.

2. The stopping station according to claim 1, wherein: the passenger boarding platform is made of at least one of concrete, asphalt, steel, steel-reinforced concrete, lawn stones, or wood.

3. The stopping station according to claim 1, wherein: the cutout is dimensioned to enable rotation of the docked wheel.

4. A method for controlling a movement of one of a plurality of vehicles to one of a plurality of vehicle stopping surfaces at a stopping station, the method comprising: receiving from the one of the plurality of the vehicles a request for stopping at the one of the plurality of the vehicle stopping surfaces of the stopping station; selecting the one of the plurality of the vehicle stopping surfaces for stopping of a requesting one of the vehicles at the one of the plurality of vehicle stopping surfaces; and transmitting the selected vehicle stopping surface to the requesting one of the plurality of the vehicles; wherein the selecting of the one of the plurality of the stopping surfaces comprises selecting cutouts for docking of wheels of the requesting one of the plurality of vehicles to a passenger boarding/alighting platform, and wherein the form of the cutout is dimensioned according to a wheel hull curve of at least one wheel of the vehicle and the number of cutouts disposed on the passenger boarding platform correlates to the number of wheels of the vehicle on a side of the vehicle facing the passenger boarding/alighting platform.

5. The method of claim 4, wherein: the selecting of the one of the plurality of the stopping surfaces is based on at least one of an expected arrival time of the requesting one of the vehicles or a number of passengers travelling in the requesting one of the vehicles.

6. The method of claim 4, wherein: the calculating is done using a structure model stored in the stopping station control unit, a known position of the requesting one of the vehicles in relation to the stopping station, and a position of the cutouts in relation to the stopping station.

7. The method of claim 4, wherein: docking of the wheels to the cutouts disposed on a passenger boarding/alighting platform adjacent to the determined one of the plurality of the vehicle stopping surfaces.

8. The method of claim 4, wherein: the transmitting of the selected vehicle stopping surfaces further comprises transmitting of conflict avoidance instructions.

9. The method of claim 4, further comprising: detecting a conflict between approaching ones of the vehicles.

10. The method of claim 9, further comprising: determining conflict avoidance instructions for avoidance of the conflict between the approaching ones of the vehicles.

12

11. A method for controlling a departure of vehicles from a stopping station in a transportation network, the method comprising:

receiving, from one of a plurality of vehicles docked to a cutout disposed on a passenger boarding/alighting platform adjacent to the one of a plurality of vehicle stopping surfaces, a request for departure from the stopping station;

determining, by a stopping station control unit, a route for leaving of the requesting one of the vehicles from the one of the vehicle stopping surfaces; and

transmitting the determined route for the leaving of the vehicle stopping surfaces to the requesting one of the plurality of vehicles.

12. The method of claim 11, wherein: the determining of the route for the leaving of the stopping station is done using a known position of a plurality of vehicles docked to the cutouts of the stopping station.

13. A method for boarding/alighting passengers at a passenger boarding/alighting platform in/from a vehicle in a transportation network, the method comprising:

approaching a vehicle stopping surface by the vehicle;

aligning wheels of the vehicle with a plurality of cutouts disposed in a vehicle-facing surface of the passenger boarding/alighting platform, wherein the form of the cutout is dimensioned according to a wheel hull curve of at least one wheel of the vehicle and the number of cutouts disposed on the passenger boarding platform correlates to the number of wheels of the vehicle on a side of the vehicle facing the passenger boarding/alighting platform;

stopping the vehicle proximate to the passenger boarding/alighting platform;

docking the wheels of the vehicle in the cutouts of the the passenger boarding/alighting platform;

boarding/alighting of passengers at the vehicle; and

leaving the stopping station by the vehicle.

14. The method according to claim 13, wherein: detecting the vehicle stopping surface disposed in the transportation network is done using an onboard positioning unit and an onboard processor of the vehicle.

15. A method for management of vehicles at a stopping station in a transportation network, the method comprising:

aligning wheels of the vehicle with a plurality of cutouts disposed at a vehicle-facing surface of the passenger boarding/alighting platform, wherein the form of the cutout is dimensioned according to a wheel hull curve of at least one wheel of the vehicle and the number of cutouts disposed on the passenger boarding platform correlates to the number of wheels of the vehicle on a side of the vehicle facing the passenger boarding/alighting platform;

stopping of the vehicle proximate to the passenger boarding/alighting platform; and

docking the wheels of the vehicle in the cutouts of the the passenger boarding/alighting platform.

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